2026 Fall

CS270: Digital Image Processing

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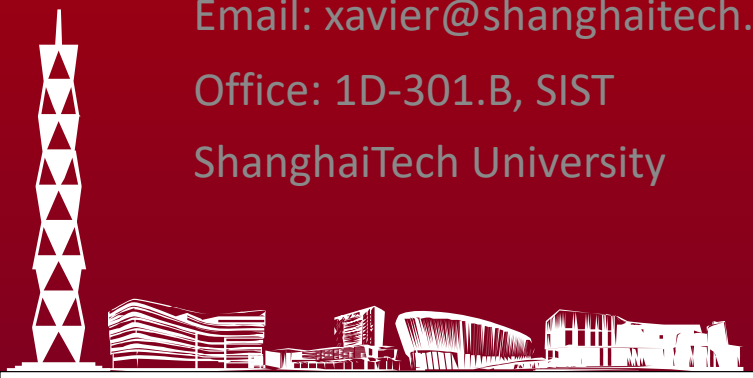
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Lecture 4
Spatial Filtering

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2026 Fall – Oct 09

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Spatial Filtering: Outline

- **Spatial filtering definition**
- **Smoothing（平滑）**
  - Linear filter
  - Non-linear filter
- **Sharpening（锐化）**
  - Spatial differentiation
  - Laplace filter

Spatial Filtering

- A Spatial filter is directly applied on the image
- A Spatial filter is also called:
  - spatial masks (掩模)
  - kernels (核)
  - templates(模板)
  - windows (窗口)
- A Spatial filter consists of:

  1) neighborhood 2) a predefined operation

- A Spatial filter can be linear and nonlinear

- Linear spatial filter corresponds to spectral filter in frequency domain
- Nonlinear spatial filter cannot be accomplished in frequency domain

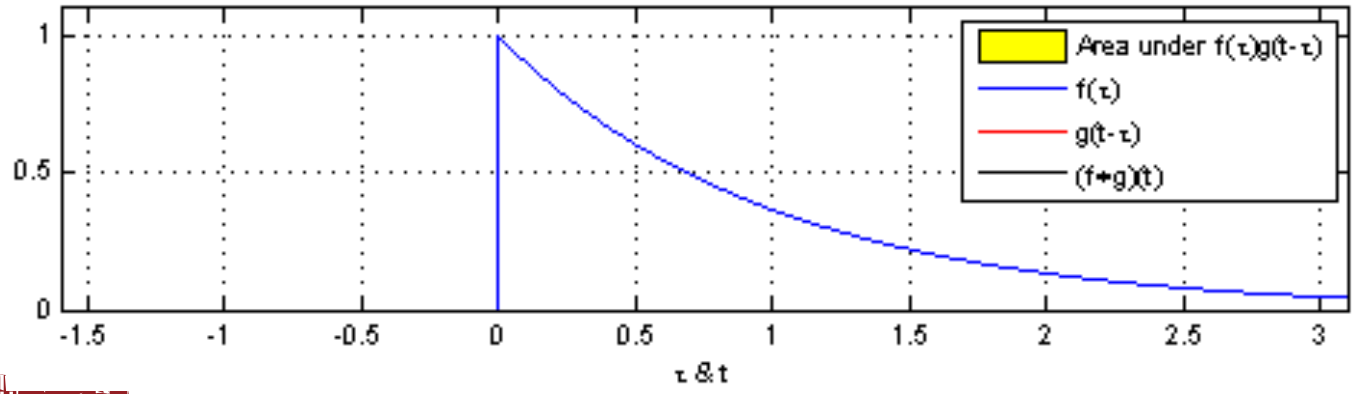
Time-domain Convolution

- Convolution of two signals $x(t)$ and $h(t)$, denoted by $x(t) * h(t)$, is defined by

$$y(t) = x(t) * h(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau$$

- For discrete-time

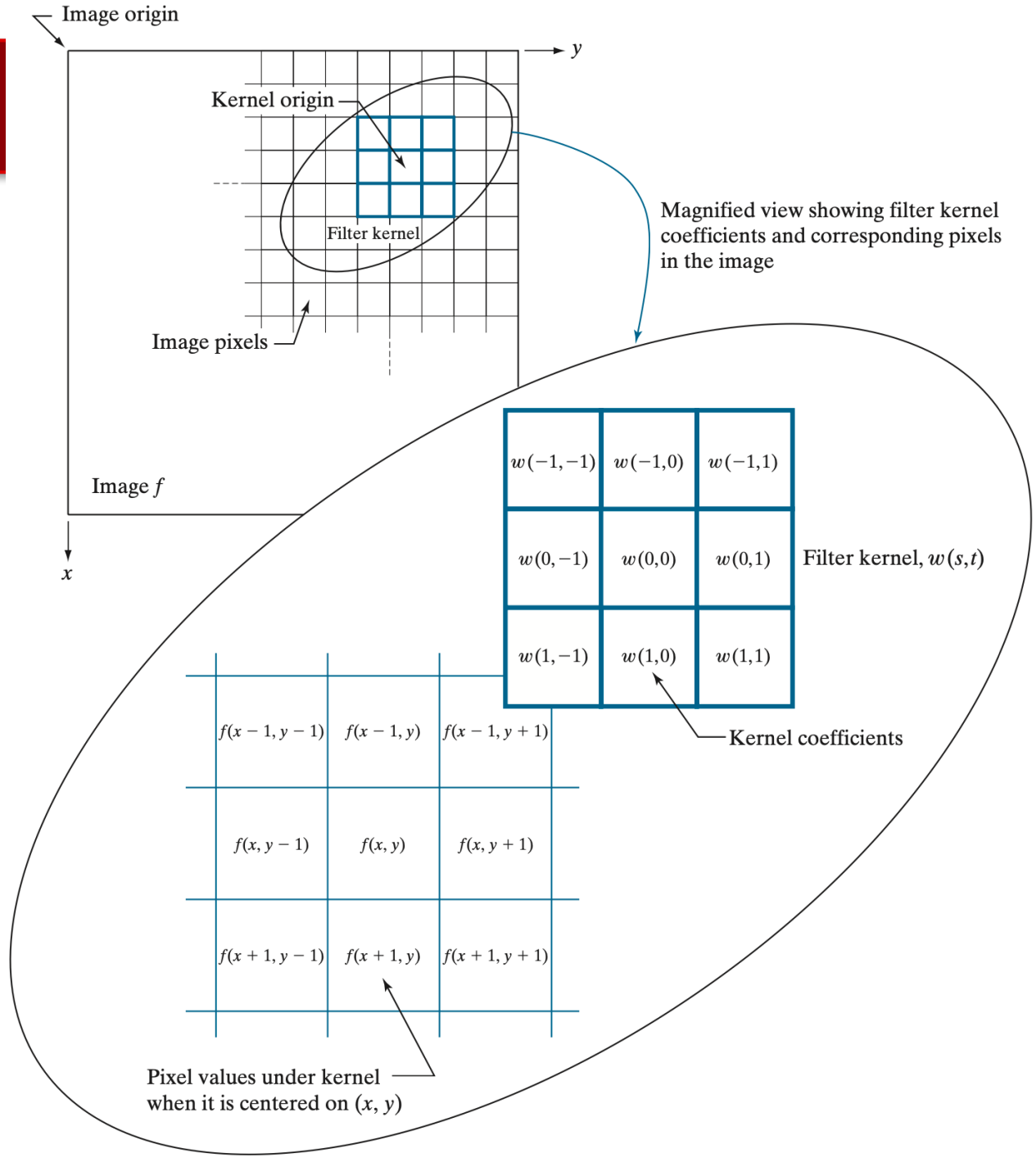
$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]$$



Spatial Filter: correlation

$$g(x,y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t) f(x+s, y+t)$$

- $f(x,y)$: input image
- $g(x,y)$: output filtered image
- $w(s,t)$: $m \times n$ spatial filter,
  where $m = 2a+1, n = 2b+1$



Spatial Filter

- For discrete-time convolution

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]$$

- Spatial filter

correlation

$$g(x,y) = \sum_{s=-a}^{a}\sum_{t=-b}^{b} w(s,t)f(x+s,y+t)$$

convolution

$$g(x,y) = \sum_{s=-a}^{a}\sum_{t=-b}^{b} w(s,t)f(x-s,y-t)$$

$w(-1,-1)$	$w(-1,0)$	$w(-1,1)$
$w(0,-1)$	$w(0,0)$	$w(0,1)$
$w(1,-1)$	$w(1,0)$	$w(1,1)$

Next: correlation or convolution ?

- What is convolution?
- What is correlation?
- What's the difference?
- How?
- Is convolution commutative?
- Is correlation commutative?
- Why convolution?
- Why correlation?







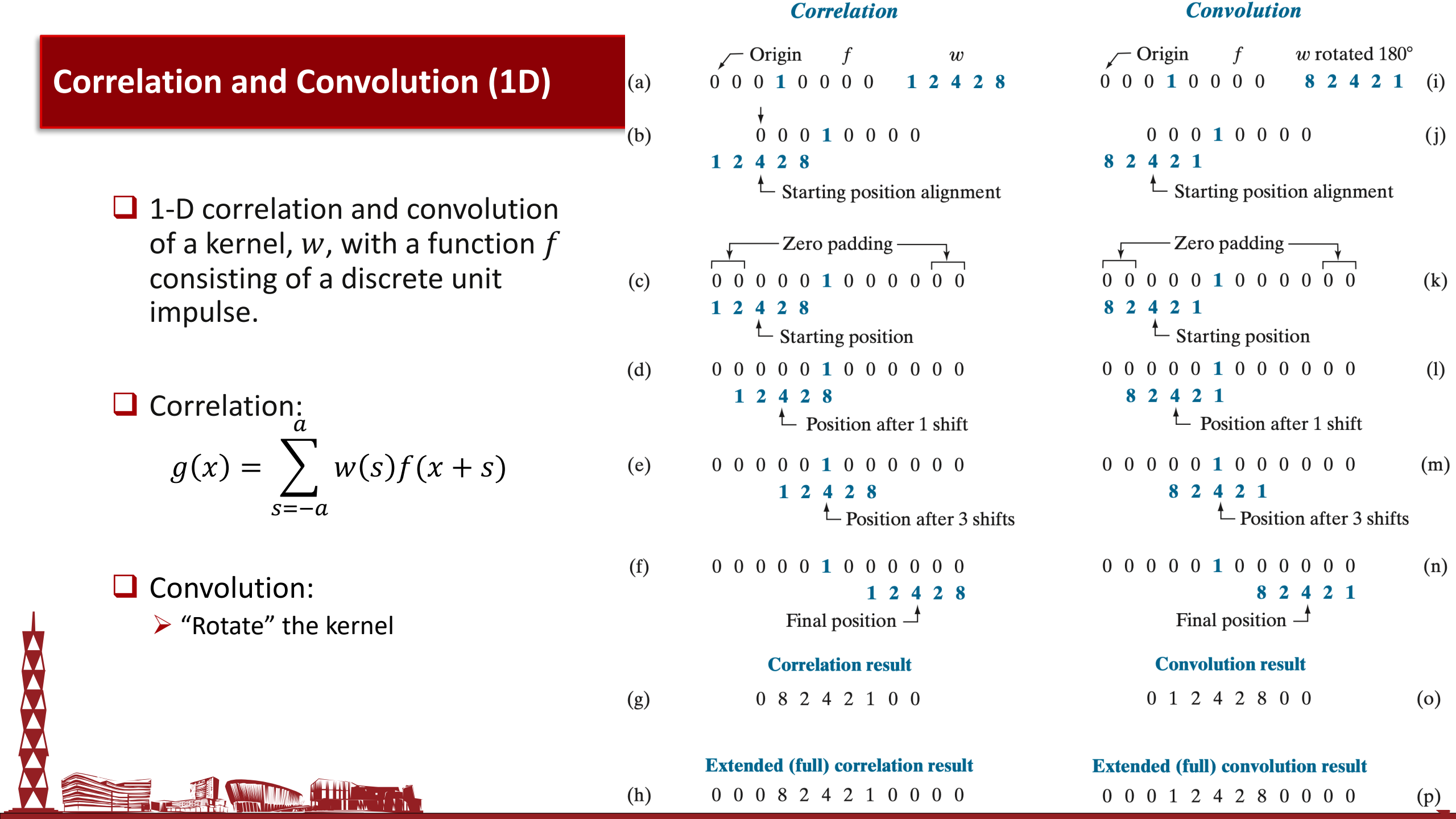
Correlation and Convolution (1D)

- 1-D correlation and convolution of a kernel, w, with a function f consisting of a discrete unit impulse.

- Correlation:

$$g(x) = \sum_{s=-a}^{a} w(s) f(x+s)$$

- Convolution:
  - "Rotate" the kernel



Correlation and Convolution (2D)

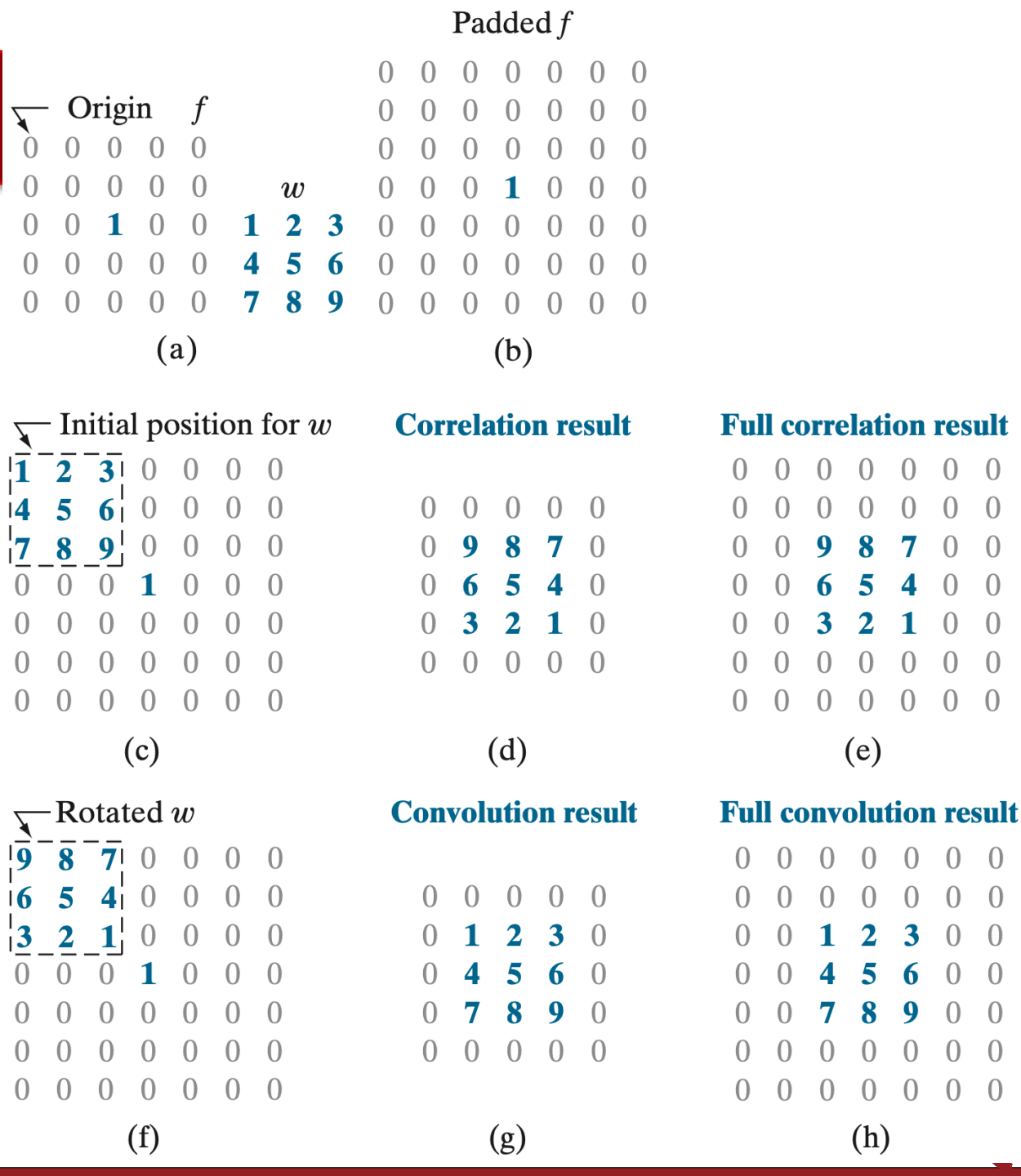
- Correlation (middle row) and convolution (last row) of a 2-D kernel with an image consisting of a discrete unit impulse.

- Correlation:

$$g(x,y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t) f(x+s, y+t)$$

- Convolution:

$$g(x,y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t) f(x-s, y-t)$$



Correlation and Convolution Equations

- **Correlation**

$$w \star f(x,y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t) f(x+s, y+t)$$

- **Convolution**

$$w \star f(x,y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t) f(x-s, y-t)$$

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

TABLE 3.5 Some fundamental properties of convolution and correlation. A dash means that the property does not hold.

Property	Convolution	Correlation
Commutative	$f \star g = g \star f$	—
Associative	$f \star (g \star h) = (f \star g) \star h$	—
Distributive	$f \star (g + h) = (f \star g) + (f \star h)$	$f \star (g + h) = (f \star g) + (f \star h)$

Spatial Filters Examples

- Linear Spatial Filter (线性滤波器)

$$R = \frac{1}{9}\sum_{k=1}^{9} z_k \text{ (Box filter)}$$

$$G(s,t) = Ke^{-\frac{s^2+t^2}{2\sigma^2}} \text{ (Gaussian filter)}$$

- Nonlinear Spatial Filter（非线性滤波器）
  - Max filter (最大值滤波)
  - Median filter (中值滤波)

Smoothing or Averaging Filters (平滑滤波器)

- Used to reduce sharp transitions in intensity (Blurring)
  - for preprocessing tasks
- Smoothing is used to reduce irrelevant detail in an image
  - pixel regions that are small with respect to the size of the filter kernel
- Noise reduction
  - Linear filter: average filtering – low-pass filter in frequency domain
  - Nonlinear filter
- Remove false contours (resulting from using an insufficient number of intensity levels)

Smoothing Filters (平滑滤波器): Box Filter

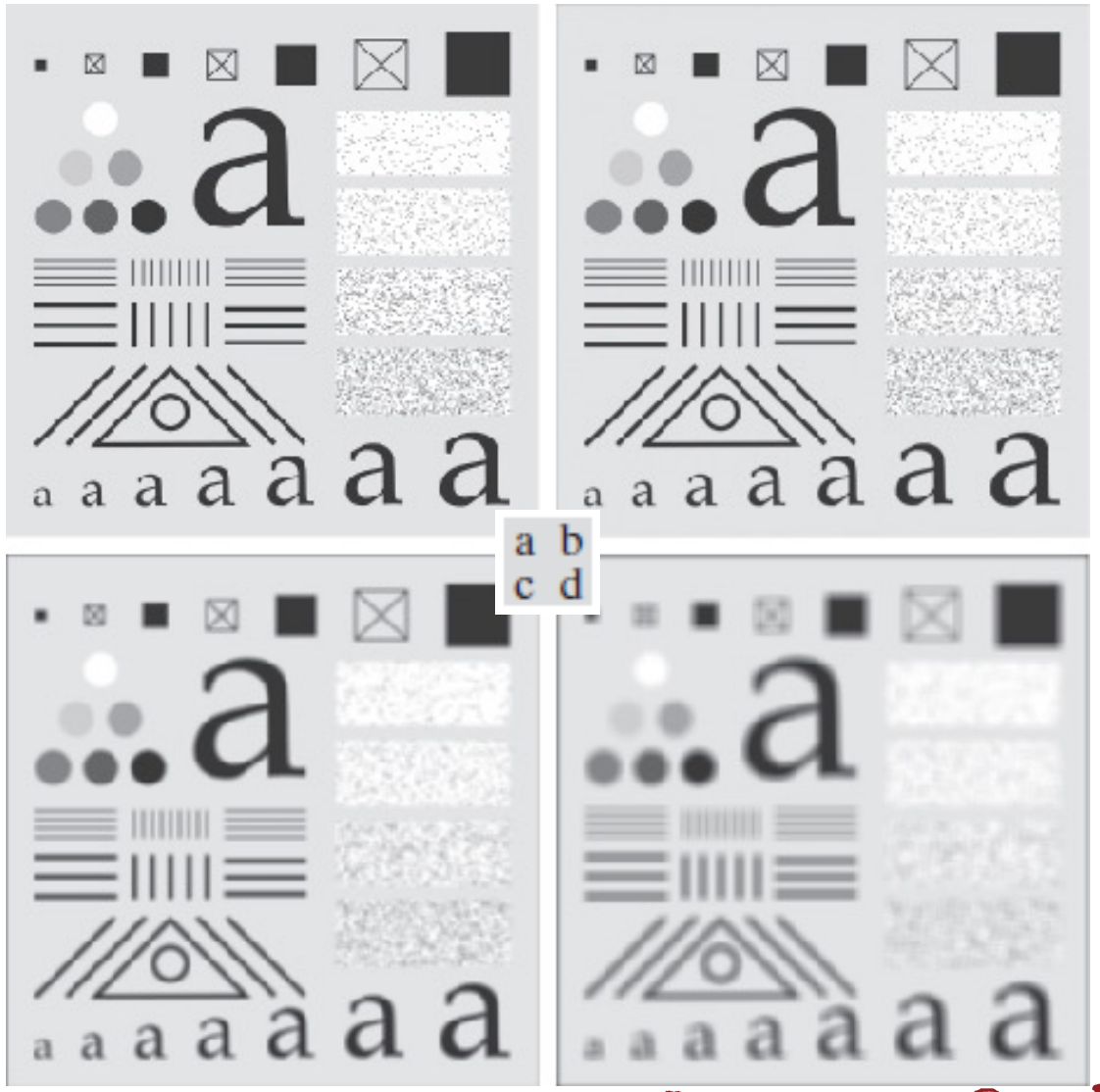
$$g(x,y) = \frac{\sum_{s=-a}^{a}\sum_{t=-b}^{b} w(s,t) f(x+s, y+t)}{\sum_{s=-a}^{a}\sum_{t=-b}^{b} w(s,t)}$$

- "Box filter" because it looks like a box when viewed in 3D
- Normalizing constant ensures that the energy remains constant when filtering
- Can use a variation when the coefficients are larger at the center
  - So pixels are more important in the averaging if closer to the center

$$\frac{1}{9} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \qquad \frac{1}{16} \times \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Impact of Filter size

- Test pattern of size 1024x1024 pixels
- Results of lowpass filtering (smoothing) with box kernels of sizes
  - (b) 3x3
  - (c) 11x11
  - (d) 21x21



a b
c d

Gaussian Filter

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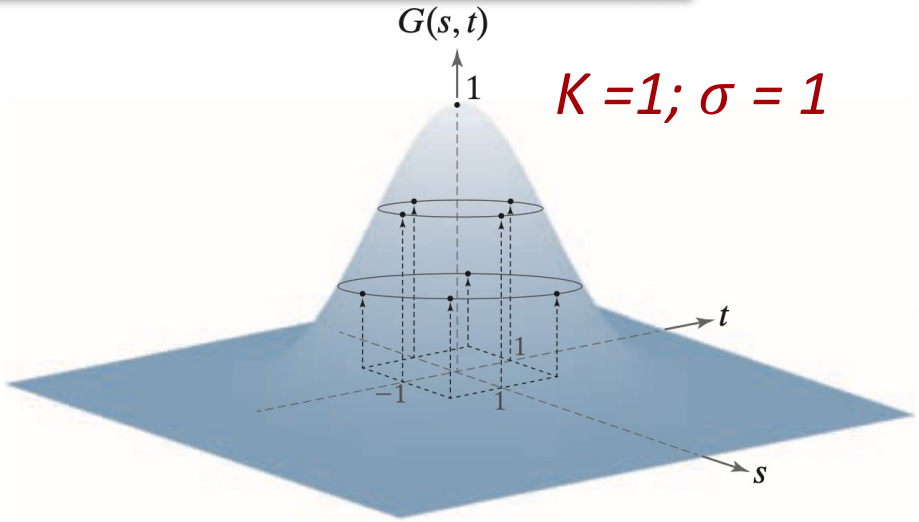
- Box filters are simple but have limits:
  - Favor blurring along image axes (artefacts)
  - Not very similar to a lens blur

- We introduce Gaussian Filters
  - *Circularly symmetric* or *isotropic*

$$G(s,t) = Ke^{-\frac{s^2+t^2}{2\sigma^2}}$$

$$G(r) = Ke^{-\frac{r^2}{2\sigma^2}}$$

$$r = \sqrt{s^2 + t^2}$$



K =1; σ = 1

$$\frac{1}{4.8976} \times \begin{bmatrix} 0.3679 & 0.6065 & 0.3679 \\ 0.6065 & 1.0000 & 0.6065 \\ 0.3679 & 0.6065 & 0.3679 \end{bmatrix}$$



a b c

FIGURE 3.42 (a)A test pattern of size 1024 × 1024. (b) Result of lowpass filtering the pattern with a Gaussian kernel of size 21 × 21, with standard deviations $\sigma = 3.5$. (c) Result of using a kernel of size 43 × 43, with $\sigma = 7$. This result is comparable to Fig. 3.39(d). We used $K = 1$ in all cases.

Gaussian Filter: a note on filter size

- The filtering effect is driven by the value of σ
  - The actual kernel size does not matter if large enough (for a given image size)

$$G(s,t) = Ke^{-\frac{s^2+t^2}{2\sigma^2}}$$

$$G(r) = Ke^{-\frac{r^2}{2\sigma^2}}$$



a b c

FIGURE 3.37 (a) Result of filtering Fig. 3.36(a) using a Gaussian kernels of size 43×43, with $\sigma = 7$. (b) Result of using a kernel of 85×85, with the same value of σ. (c) Difference image.

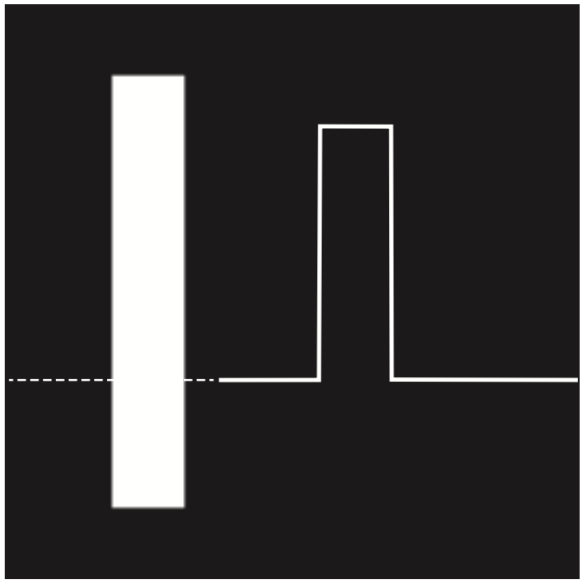
Box VS Gaussian



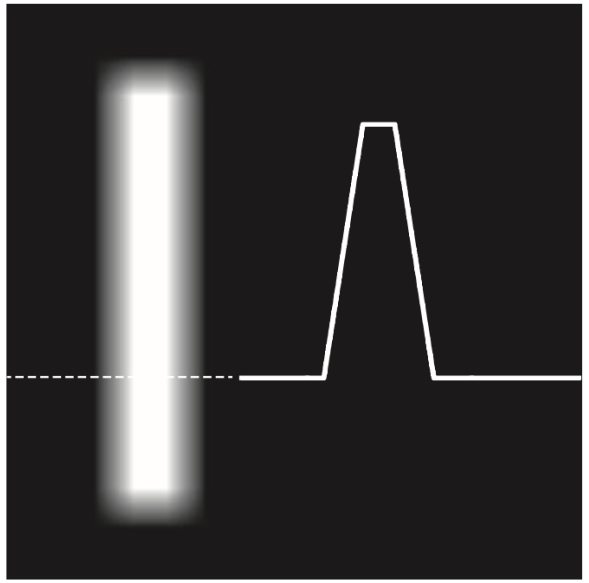
Box Filter



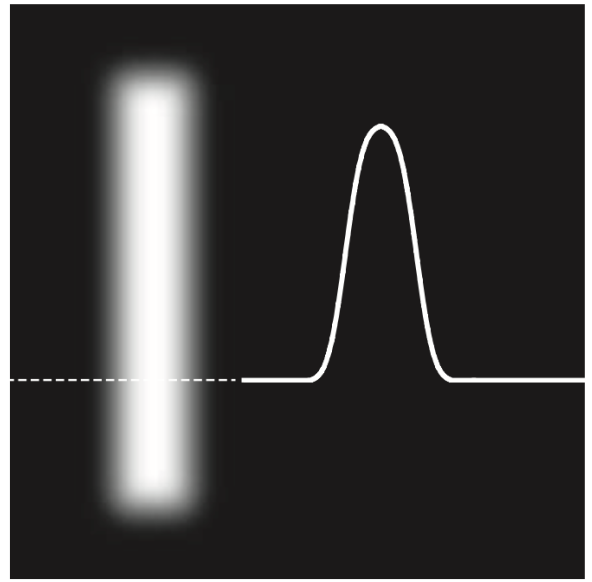
Gaussian Filter



Original image
And
intensity profile



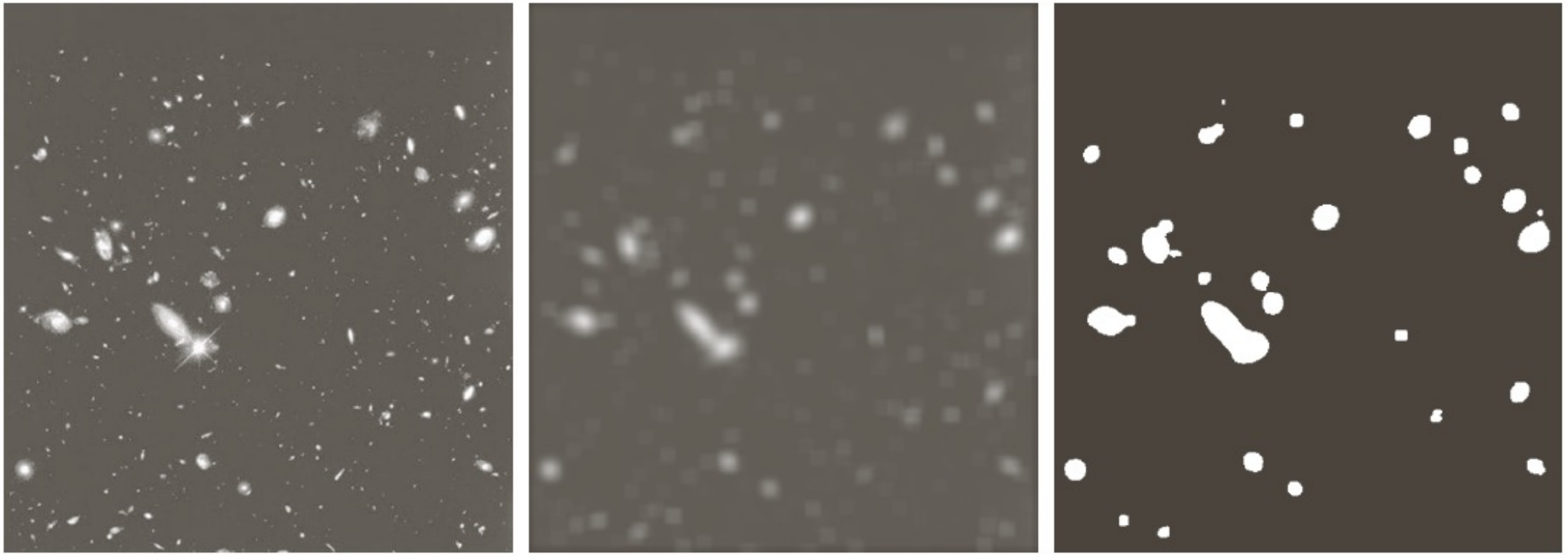
Box kernel
Size 71x71



Gaussian kernel
Size 151x151
K=1, σ=25

- Transitions with Gaussian filter are much smoother
  - Box Filter: linear smoothing, ramps
  - Gaussian Filter: smoother results around the edge transitions

Application example: Smoothing + Thresholding(阈值处理)



- Use gaussian filter to emphasize large bright regions
- Threshold intensity level to remove *irrelevant* details (i.e. too small)

Application example: Shading correction



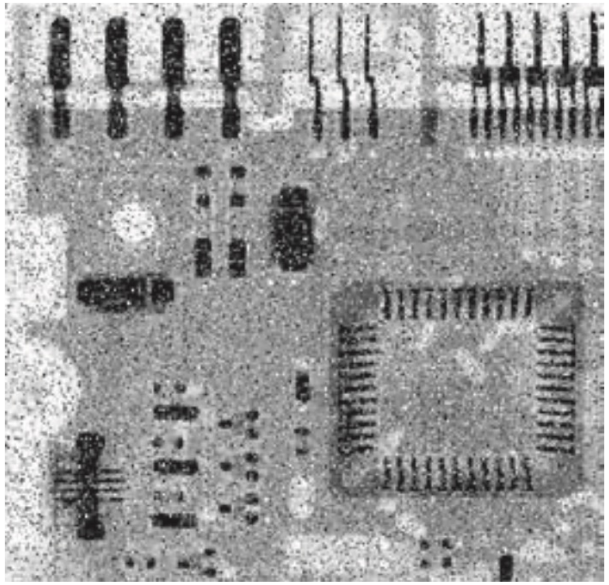
- Image shaded by a shading pattern oriented in the -45° direction
- Estimate the shading patterns obtained using lowpass (smoothing filtering): very large kernel
- Divide the original image by the estimated shading pattern

Nonlinear Smoothing Filters

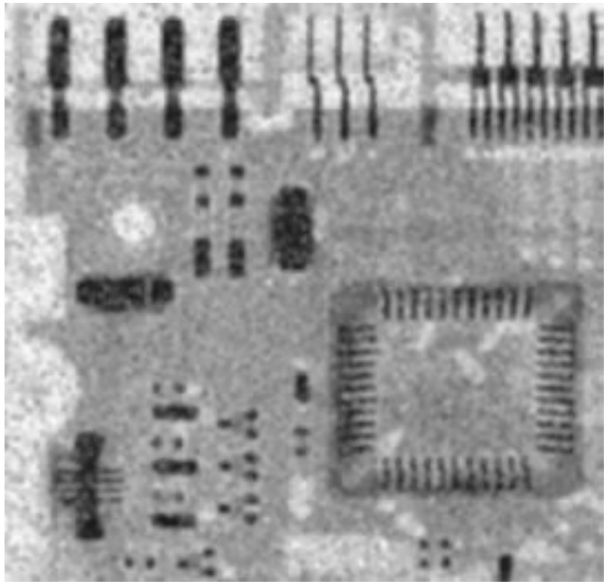
- Order-statistic filter (统计排序滤波器): based on ranking pixels in neighborhood
  - Ex： median filter （中值滤波器）

$$g(x,y) = median\{m \times n \text{ pixel neighboring around } I(x,y)\}$$

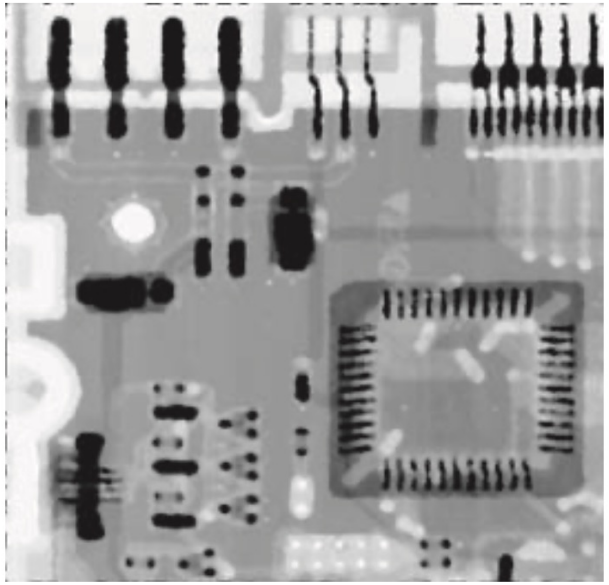
50	48	46	42
52	0	50	48
46	47	55	40
51	48	46	42



Original X-ray image
Salt-and-pepper noise



Gaussian filtering
Kernel: 19x19, σ=3



Median filtering
Kernel: 7x7

Sharpening Filters

- Spatial differentiation（空间微分）
- Sharpening filter
  - Laplacian filtering（拉普拉斯算子）

Sharpening Filters

- To highlight transitions in intensity
- Accomplished by spatial differentiation
  - First-order derivative: $\frac{\partial f}{\partial x} = f(x+1) - f(x)$
  - Second-order derivative: $\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$

Laplacian(拉普拉斯算子)

- For an image function $f(x, y)$:
  - X direction: $\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$
  - Y direction: $\frac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$

$$\nabla^2 f(x, y) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$ Discrete Laplacian

$$= f(x, y+1) + f(x, y-1) + f(x+1, y) + f(x-1, y) - 4f(x, y)$$

Laplacian Filter Masks

- Can we write the Laplacian as a filter kernel ?

$$\nabla^2 f(x,y) = f(x,y+1) + f(x,y-1) + f(x+1,y) + f(x-1,y) - 4f(x,y)$$

0	1	0
1	−4	1
0	1	0

1	1	1
1	−8	1
1	1	1

0	−1	0
−1	4	−1
0	−1	0

−1	−1	−1
−1	8	−1
−1	−1	−1

$$\begin{aligned}\nabla^2 f(x,y) = {} & f(x,y+1) + f(x,y-1) + f(x+1,y) + f(x-1,y) \\ & + f(x+1,y+1) + f(x+1,y-1) + f(x-1,y-1) + f(x-1,y+1) \\ & - 8f(x,y)\end{aligned}$$

Laplacian Filter Masks – Sharpening filter

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- Laplacian is a derivative operator:
  - highlights sharp intensity transitions in an image
  - de-emphasizes regions of slowly varying intensities
- Grayish edge lines and other discontinuities + dark, featureless background
- Add back the original image to recover background

	1	

$-$

	1	
1	-4	1
	1	

$=$

	-1	
-1	5	-1
	-1	

	1	

$+$

	-1	
-1	4	-1
	-1	

$=$

	-1	
-1	5	-1
	-1	

$$g(x,y) = f(x,y) + c\nabla^2 f(x,y), \qquad \text{where } c = \pm 1$$

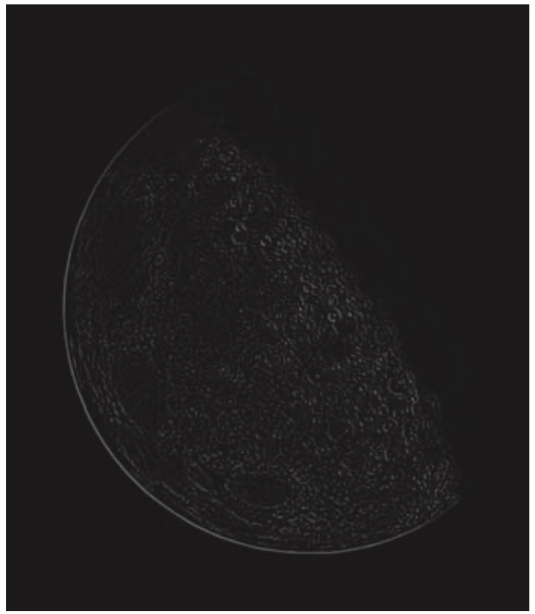


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Image Sharpening with Laplacian

Blurred image of the North Pole of the moon





Laplacian image obtained with ”simple” kernel



Image sharpened using previous equation

0	1	0
1	−4	1
0	1	0





Same procedure but with “diagonal” Laplacian kernel

1	1	1
1	−8	1
1	1	1

Implementations in Matlab

Low pass filter example:

```
>> LP = 1/9 *[1,1,1;1,1,1;1,1,1];

>> im3 = imfilter(com,LP);

>>figure; imshow(im3,[]);
```

Sharpening filter example:

```
>> f3 = [-1,-1,-1; -1,8,-1;-1,-1,-1];

>> J1 = imfilter(im,f3);

>>figure; imshow(J1,[]);
```

Median filter example:

```
>> J2 = medfilt2(im,[3 3]);

>> J4 = medfilt2(im,[6 6]);

>> J3 = medfilt2(im,[11 1]);

>> J5 = medfilt2(im,[1 11]);
```



Take home message

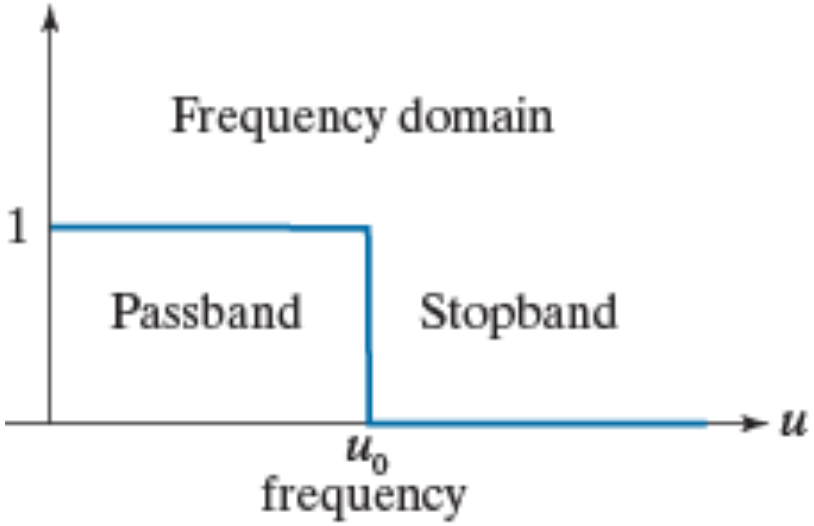
- For image processing, the spatial domain processing is similar to 1-D signal processing in time domain.
- The spatial filter we mentioned in this lecture is actually correlation between image and the filter, however, when using a filter symmetric around the center, it is equivalent to convolution.
- “convolving a kernel with an image” is often used to denote the sliding, sum-of-products process and does not necessarily differentiate between correlation and convolution. Rather, it is used generically to denote either of the two operations. This imprecise terminology is a frequent source of confusion
- Common spatial filters involving smoothing filter and sharpening filter.

Other commonly used spatial filters

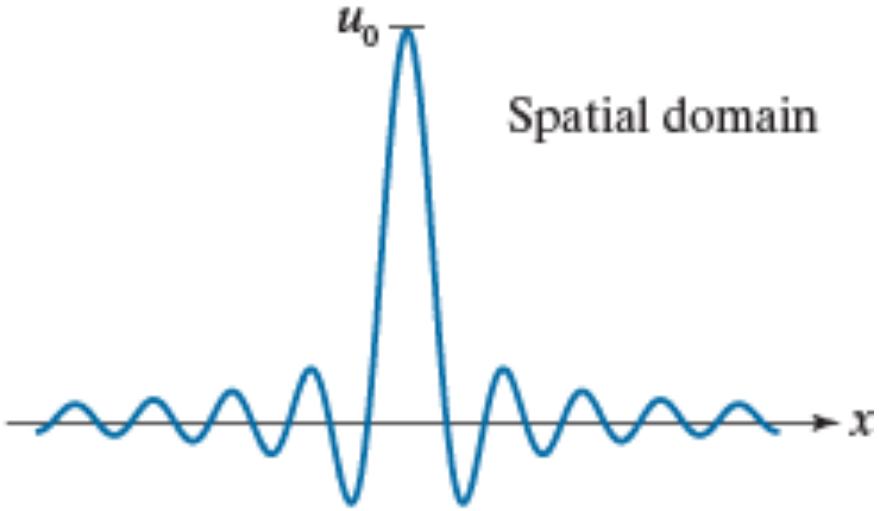
- Some other perspectives on spatial filtering

- Sobel Filter
- Unsharp Masking (非锐化掩蔽)
- LoG Filter (Laplacian of Gaussian)
  - useful for finding edges
  - also useful for finding blobs

Filtering in frequency domain and spatial domain



Ideal 1D Low-Pass filter



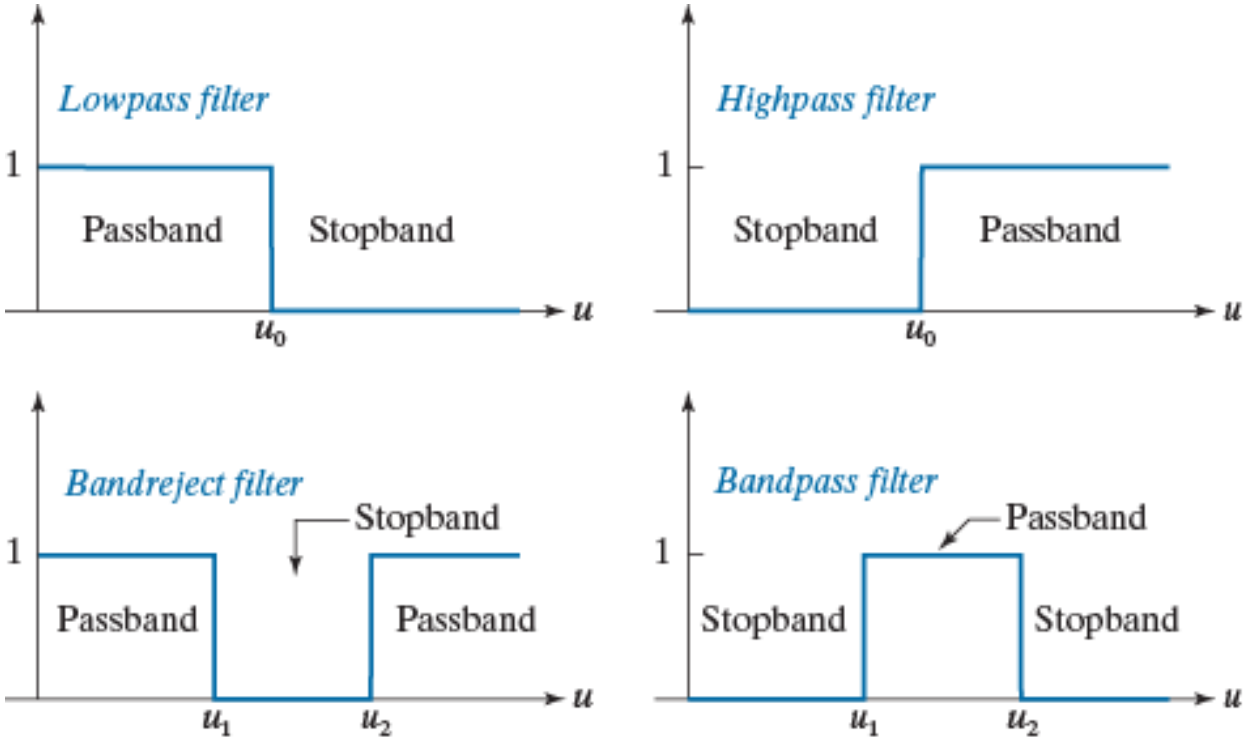
Ideal 1D Low-Pass filter kernel in spatial domain

Q: Is ideal filter really ideal for image processing?

Filtering in frequency domain and spatial domain

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- 4 types of filters

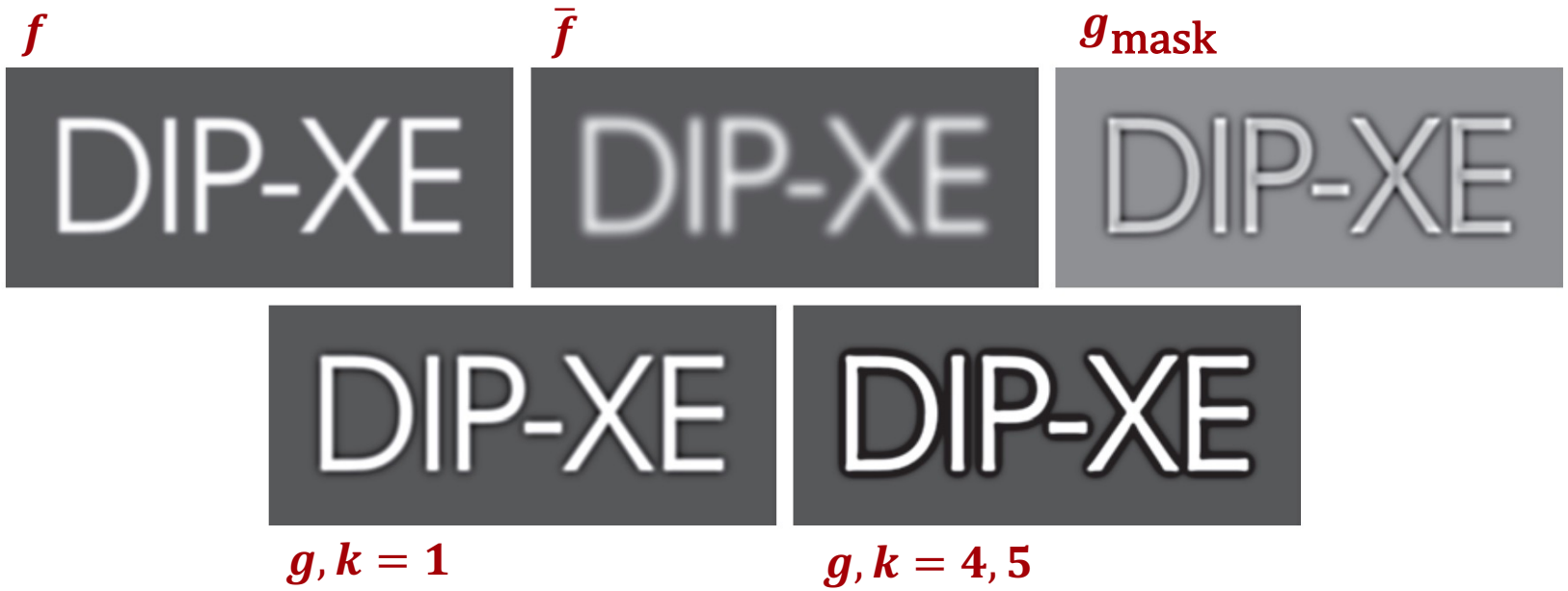


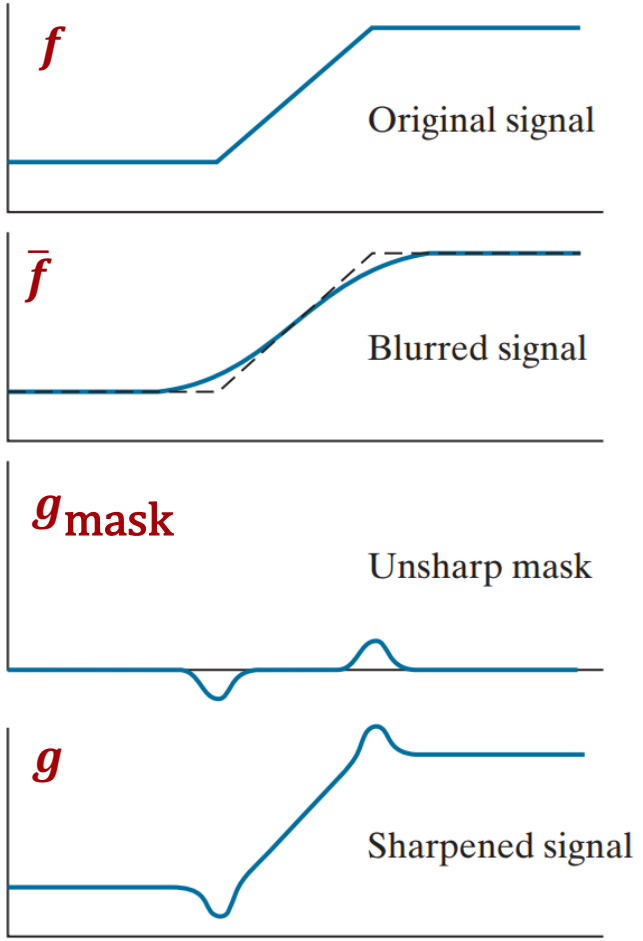
Filter type	Spatial kernel in terms of lowpass kernel, *lp*
Lowpass	$lp(x,y)$
Highpass	$hp(x,y) = \delta(x,y) - lp(x,y)$
Bandreject	$br(x,y) = lp_1(x,y) + hp_2(x,y)$ $= lp_1(x,y) + [\delta(x,y) - lp_2(x,y)]$
Bandpass	$bp(x,y) = \delta(x,y) - br(x,y)$ $= \delta(x,y) - [lp_1(x,y) + [\delta(x,y) - lp_2(x,y)]]$

Unsharp Mask (非锐化掩蔽)

$$g_{\text{mask}}(x,y) = f(x,y) - \bar{f}(x,y)$$

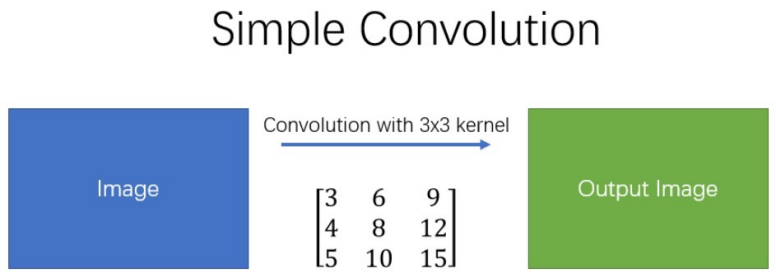
$$g(x,y) = f(x,y) + k * g_{\text{mask}}(x,y)$$

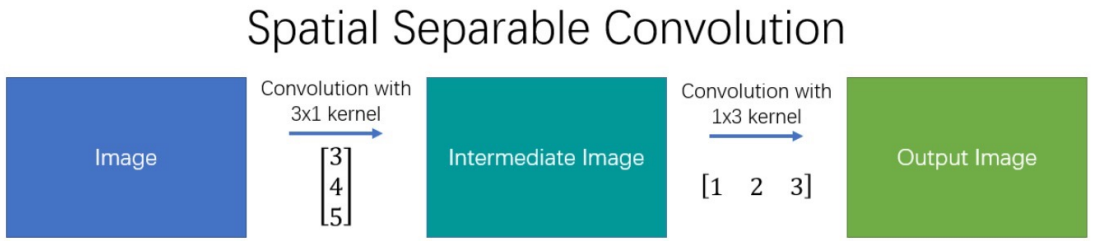




Separable filter kernels

- Example: $\begin{bmatrix} 3 & 6 & 9 \\ 4 & 8 & 12 \\ 5 & 10 & 15 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$
- $\boldsymbol{w = ab^T}$: outer product of 2 vectors equivalent to convolution !
- If $\boldsymbol{w = w_1 \star w_2}$, then
  - $\boldsymbol{w \star f = (w_1 \star w_2) \star f = (w_2 \star w_1) \star f = w_2 \star (w_1 \star f) = (w_1 \star f) \star w_2}$



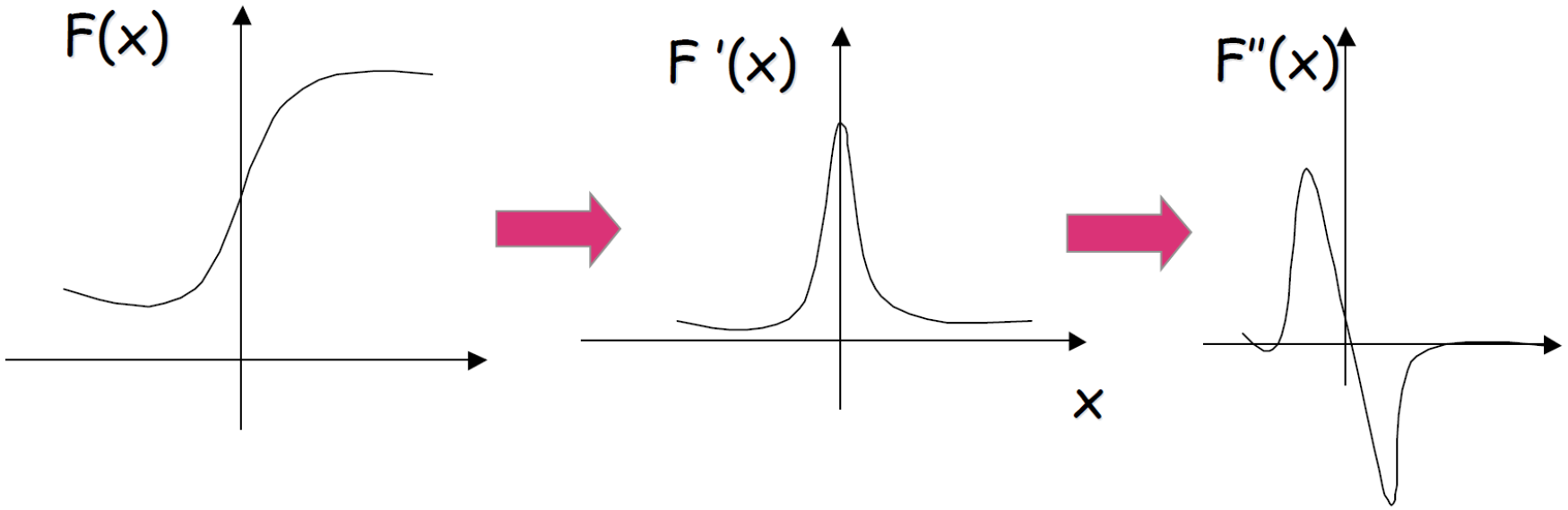


Computational advantage:

$$C = \frac{MNmn}{MN(m+n)} = \frac{mn}{(m+n)}$$

Recall: First & Second-Derivative filters

- Sharp changes in gray level of the input image corresponds to “peaks or valleys” of the first-derivative of the input signal.
- Peaks or valleys of the first derivative of the input signal, correspond to “zero-crossings” of the second-derivative of the input signal.



Laplacian (拉普拉斯算子)

- For an image function $f(x, y)$:
  - X direction: $\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$
  - Y direction: $\frac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$

$$\nabla^2 f(x, y) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$ Discrete Laplacian

$$= f(x, y+1) + f(x, y-1) + f(x+1, y) + f(x-1, y) - 4f(x, y)$$

Laplacian Filter Masks

- Laplacian as a filter kernel

$$\nabla^2 f(x,y) = f(x,y+1) + f(x,y-1) + f(x+1,y) + f(x-1,y) - 4f(x,y)$$

0	1	0
1	−4	1
0	1	0

1	1	1
1	−8	1
1	1	1

0	−1	0
−1	4	−1
0	−1	0

−1	−1	−1
−1	8	−1
−1	−1	−1

$$\begin{aligned}\nabla^2 f(x,y) = {} & f(x,y+1) + f(x,y-1) + f(x+1,y) + f(x-1,y) \\ & + f(x+1,y+1) + f(x+1,y-1) + f(x-1,y-1) + f(x-1,y+1) \\ & - 8f(x,y)\end{aligned}$$

Laplacian (拉普拉斯算子)

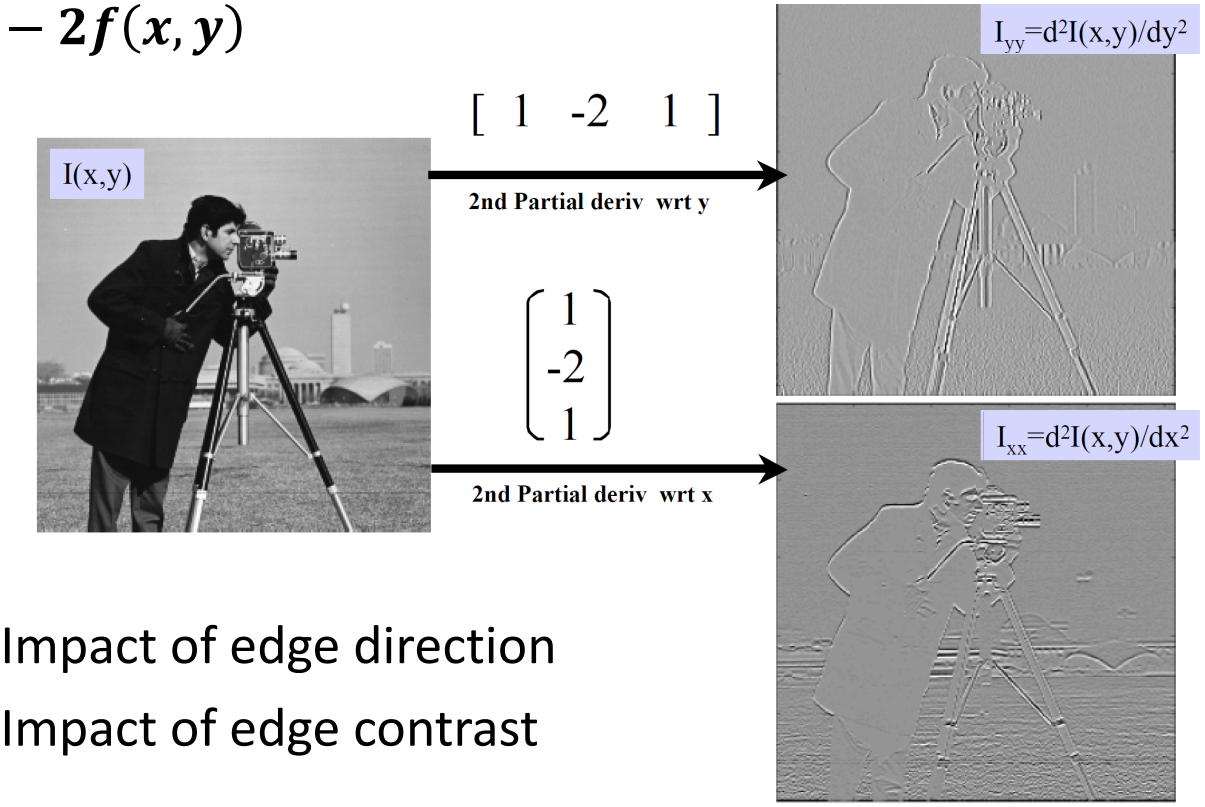
- For an image function $f(x, y)$,
  - X direction: $\frac{\partial 2f}{\partial x2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$
  - Y direction: $\frac{\partial 2f}{\partial y2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$

I_{yy}

1	-2	1

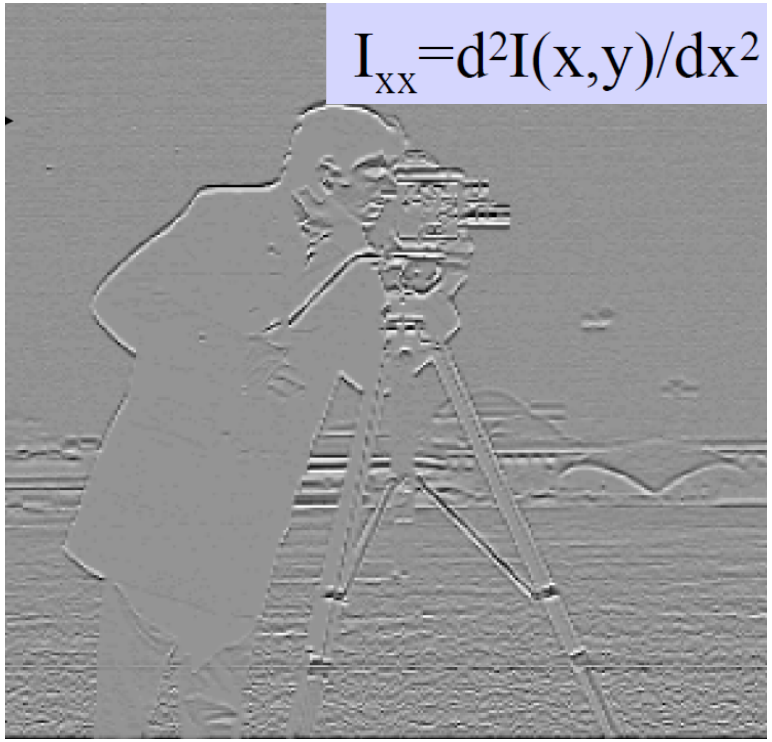
1
-2
1

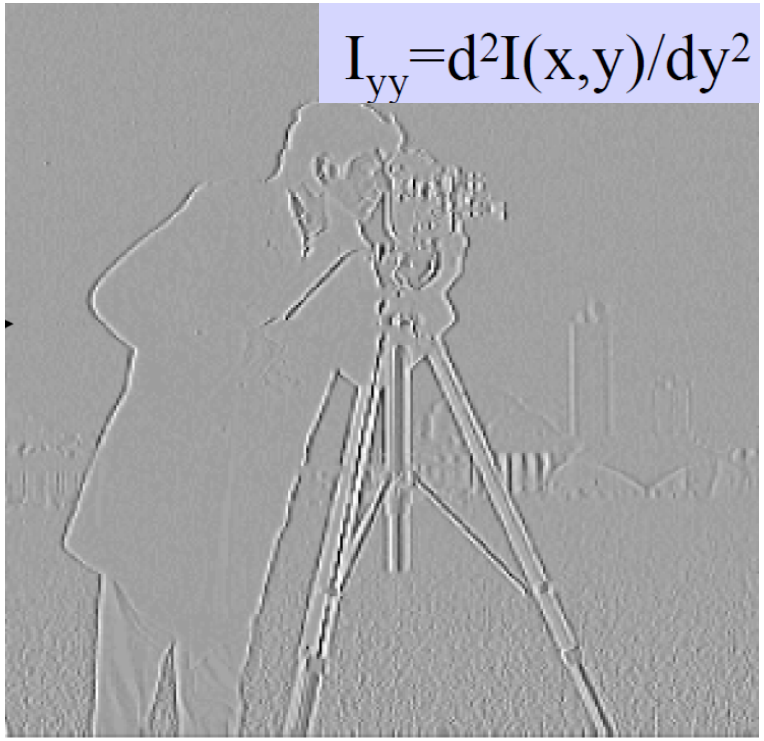
I_{xx}



1. Impact of edge direction
2. Impact of edge contrast

Laplacian







Gradient (梯度)

The first-order derivative of $f(x, y)$: $\nabla \boldsymbol{f} \equiv \mathbf{grad}(\boldsymbol{f}) \equiv \begin{cases} \boldsymbol{g_x} \\ \boldsymbol{g_y} \end{cases} = \begin{cases} \dfrac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} \\ \dfrac{\partial \boldsymbol{f}}{\partial \boldsymbol{y}} \end{cases}$

points in the direction of the greatest rate of change of f at location (x, y)

The amplitude: $\boldsymbol{M(x, y)} = \mathbf{mag}(\nabla \boldsymbol{f}) = \sqrt{\boldsymbol{g_x^2} + \boldsymbol{g_y^2}}$

$$\boldsymbol{M(x, y)} \approx |\boldsymbol{g_x}| + |\boldsymbol{g_y}|$$

To help computation complexity !
But keeps the main properties of the amplitude

Gradient (梯度)

- Simplest possibility
  - $g_x = (z_8 - z_5)$ and $g_y = (z_6 - z_5)$

- Roberts cross-gradient operator（罗伯特交叉梯度算子）
  - Uses cross differences
  - $\boldsymbol{M(x,y) \approx |g_x| + |g_y|}$
  - $\boldsymbol{= |z_9 - z_5| + |z_8 - z_6|}$

−1	0
0	1

0	−1
1	0

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

- But the kernel does not have an odd size, we prefer at least a 3x3 kernel…

Gradient (梯度)

上海科技大学
ShanghaiTech University

- Sobel operator （Sobel算子）
- $M(x,y) = |(z_7+2z_8+z_9)-(z_1+2z_2+z_3)| + |(z_3+2z_6+z_9)-(z_1+2z_4+z_7)|$

g_x g_y

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

−1	−2	−1
0	0	0
1	2	1

g_x

−1	0	1
−2	0	2
−1	0	1

g_y

Averaging

Partial derivative

Q: How to really understand Sobel operator? What are the functions?

Sobel operator



g_x

−1	−2	−1
0	0	0
1	2	1

Horizontal edges



g_y

−1	0	1
−2	0	2
−1	0	1

Vertical edges





Sobel operator



Notes about the Laplacian

- $\nabla^2 I(x,y)$ is a SCALAR
  - ✅ Can be found using a SINGLE mask
  - ❌ Orientation information is lost
- $\nabla^2 I(x,y)$ is the sum of SECOND-order derivatives
  - But taking derivatives increases noise.
  - Very noise sensitive!
- It is always combined with a smoothing operation.

Laplacian of Gaussian (LoG) Filter

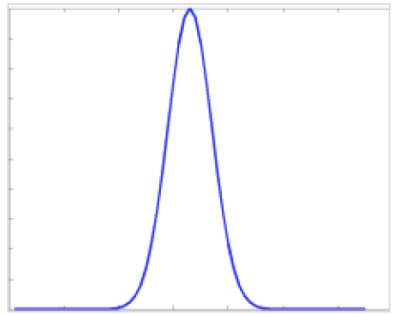
- First smooth (Gaussian filter),
- Then, second order derivative (Laplacian filter):

$$\nabla^2 (G(x,y))$$

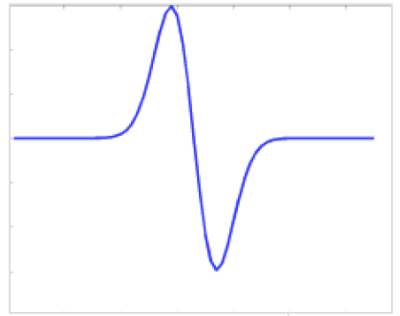
$$G(x,y) = e^{-\frac{x^2+y^2}{2\sigma^2}}$$

$$G'(x,y) = -\frac{1}{2\sigma^2} 2(x+y) e^{-\frac{x^2+y^2}{2\sigma^2}} = -\frac{x+y}{\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

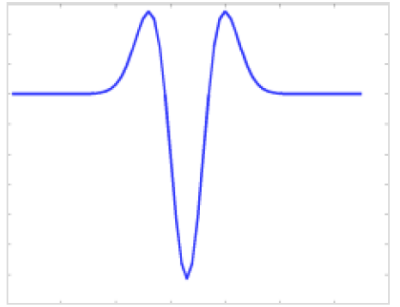
$$G''(x,y) = -\frac{1}{\pi\sigma^4}\left(1-\frac{x^2+y^2}{2\sigma^2}\right) e^{-\frac{x^2+y^2}{2\sigma^2}}$$



$G(x,y)$



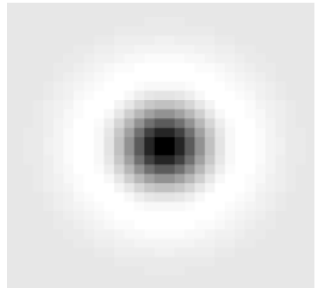
$G'(x,y)$

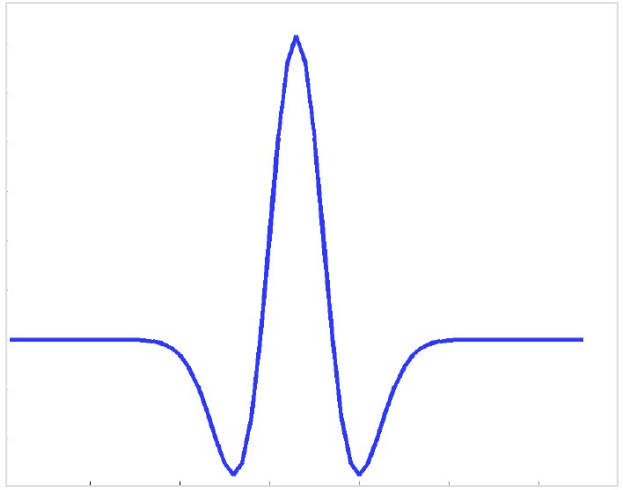


$G''(x,y)$

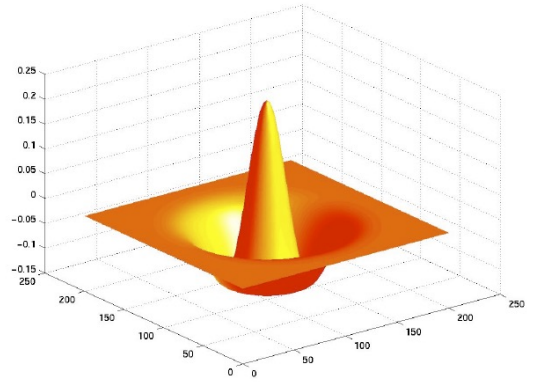
Second derivative of a Gaussian

$$G''(x,y) = -\frac{1}{\pi\sigma^4}\left(1-\frac{x^2+y^2}{2\sigma^2}\right)e^{-\frac{x^2+y^2}{2\sigma^2}}$$





2D analog



LoG "Mexican Hat"

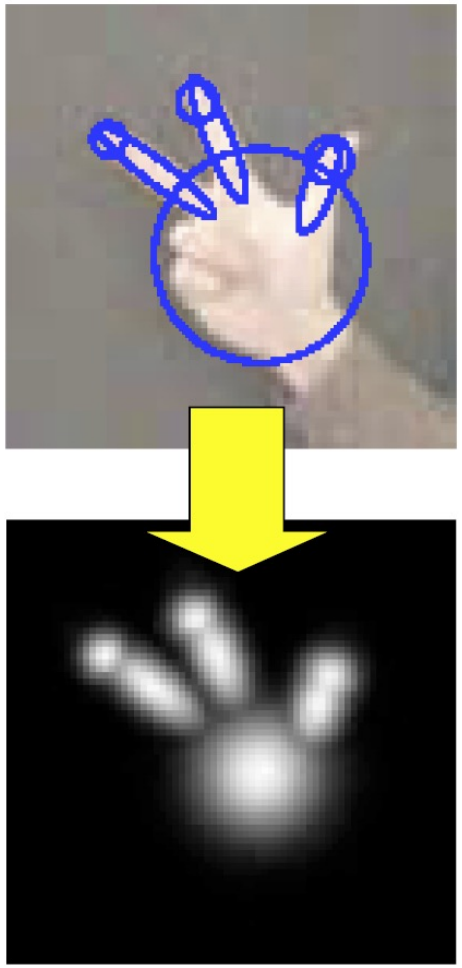
Effect of LoG Filter

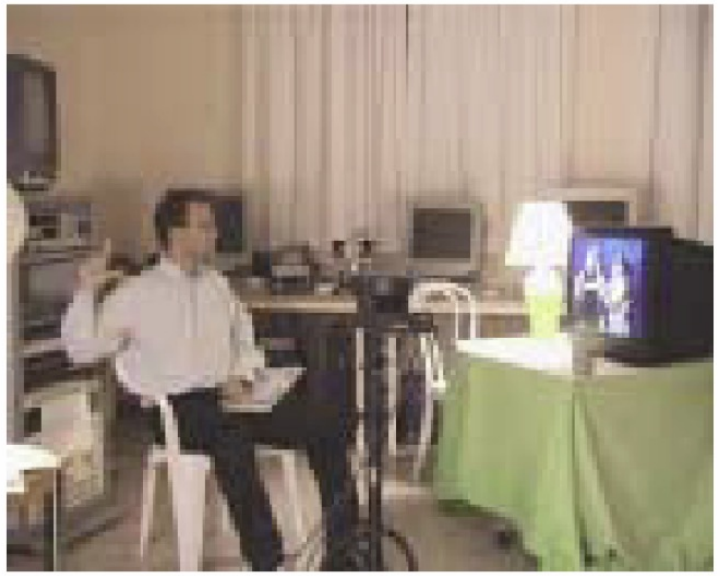
Sigma = 1 Sigma = 4 Sigma = 10



Band-Pass Filter (suppresses both high and low frequencies)

Application of LoG Filter





Gesture recognition for
the ultimate couch potato

Matlab practice: spatial filtering

- w = fspecial('type', parameters)
- g = imfilter(f, w, 'replicate')

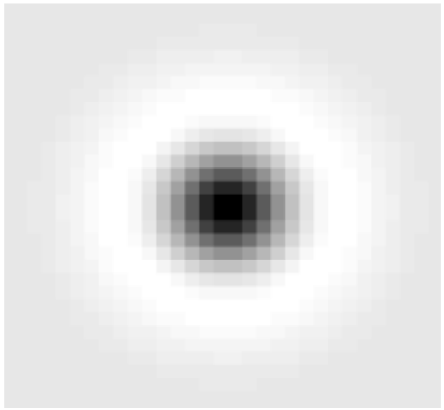
- See some examples.

- Then practice by yourself…

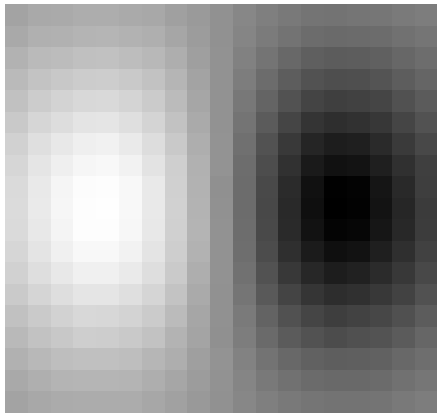
Take home message

- Key idea: Cross correlation with a filter can be viewed as comparing a little “picture” of what you want to find against all local regions in the image.

LoG



Derivative of Gaussian



Take home message

